Loan Approval Prediction Using Machine Learning

Internship: InternSpark Data Science Internship
Task: Task 2 — Loan Approval Prediction
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1. Objective

The objective of this project is to develop supervised machine learning models that predict loan approval outcomes using applicant information. The project focuses on data preprocessing, class imbalance, model comparison, and evaluation metrics.

2. Dataset Overview

The dataset contains 614 loan applications and 13 columns, including the loan identifier, applicant characteristics, financial information, and loan approval status.

The target variable is Loan_Status:

- Y: Loan approved
- N: Loan not approved

The dataset contains 422 approved applications and 192 rejected applications. Seven columns contain missing values, and no duplicate rows were found.

3. Data Preprocessing

The following preprocessing steps were implemented:

- Removed Loan_ID from the model input features because it is an identifier.
- Separated the input features from the target variable.
- Used median imputation for missing numerical values.
- Used most-frequent-value imputation for missing categorical values.
- Applied standard scaling to numerical features.
- Applied one-hot encoding to categorical features.
- Split the dataset into 80% training data and 20% testing data using stratified sampling.

Class weights were used to account for the imbalance between approved and rejected applications.

4. Machine Learning Models

Two supervised classification algorithms were evaluated:

1. Logistic Regression
2. Random Forest Classifier

Both models were implemented using pipelines that combine preprocessing and classification.

5. Model Evaluation

The reported evaluation results are:

Metric	Logistic Regression	Random Forest
Accuracy	0.829	0.829
Precision	0.872	0.840
Recall	0.882	0.929
F1-score	0.877	0.883
ROC-AUC	0.855	0.773

Both models achieved 82.9% accuracy.

Logistic Regression achieved higher precision and ROC-AUC in the reported results. Random Forest achieved higher recall and F1-score.

The Random Forest model is a candidate for further evaluation when recall and F1-score are priorities. However, the final model should be selected after reviewing the confusion matrices, verifying the reported metrics, and evaluating the business consequences of incorrect predictions.

6. Business Interpretation

A loan prediction model can help financial institutions prioritize applications for further review. However, incorrect predictions may have important consequences.

- False approvals may expose a lender to financial risk.
- False rejections may unfairly exclude eligible applicants from credit.

The model should support human review rather than make automatic lending decisions. Before deployment, the institution should assess fairness, data quality, model calibration, and performance across relevant applicant groups.

7. Threshold Selection

The default classification threshold is generally 0.50. A lower threshold may increase the number of applications predicted as approved, while a higher threshold may reduce them.

The deployment threshold should be selected using validation data, expected financial costs, customer impact, and fairness requirements. The threshold should not be chosen solely to maximize accuracy.

8. Limitations

The dataset contains only 614 records, so the reported results may vary with different data samples. The evaluation results should be verified, and further testing on independent data is needed before real-world use.

Class weighting was applied, but additional approaches such as SMOTE or undersampling have not yet been evaluated in this report.

9. Conclusion

This project demonstrates a complete initial machine learning workflow for loan approval prediction, including data inspection, missing-value handling, categorical encoding, numerical scaling, model training, and evaluation.

Both Logistic Regression and Random Forest achieved 82.9% reported accuracy. Random Forest achieved a higher reported F1-score, while Logistic Regression achieved a higher reported ROC-AUC. Further validation and business analysis are necessary before selecting a final deployment model.